

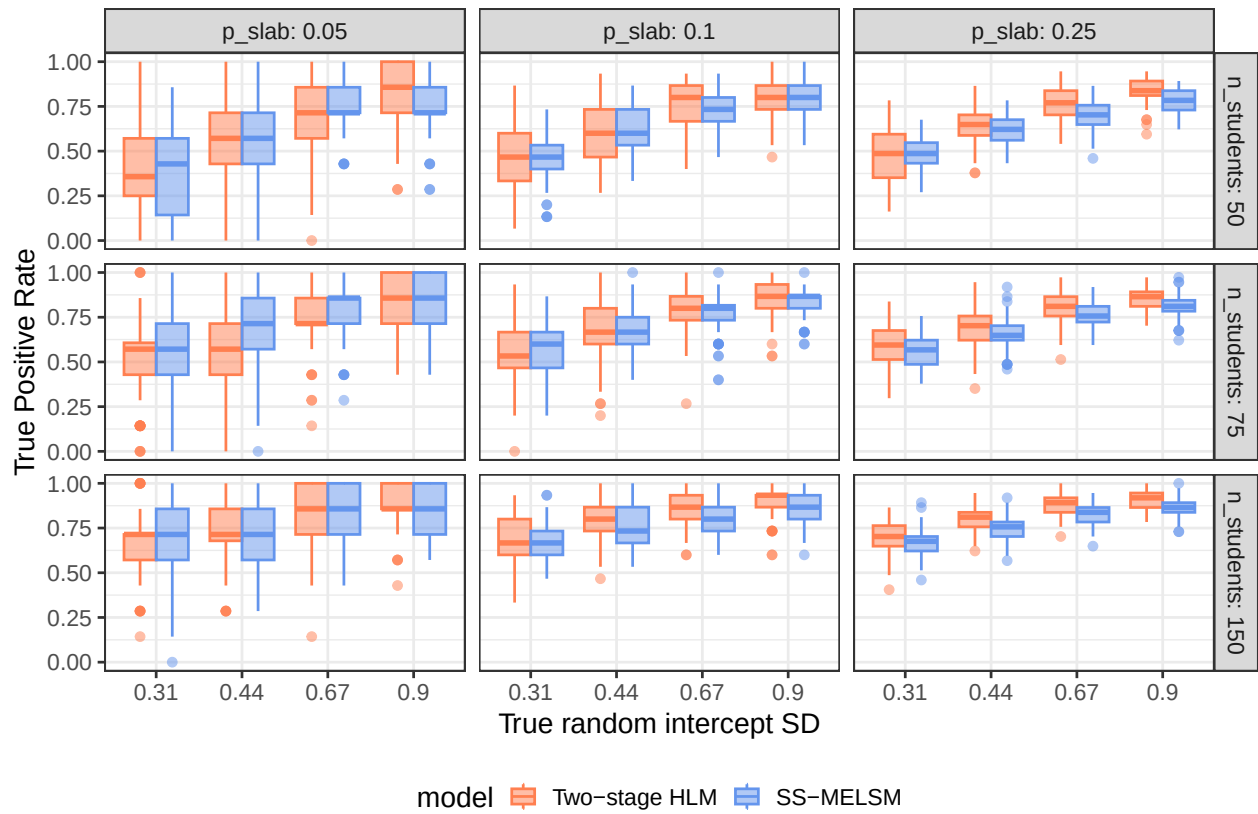
SS-MELSM vs Two-stage HLM

Study 1: Main simulation

Uncorrelated random effects ($\rho_{u_0t_0} = 0$)

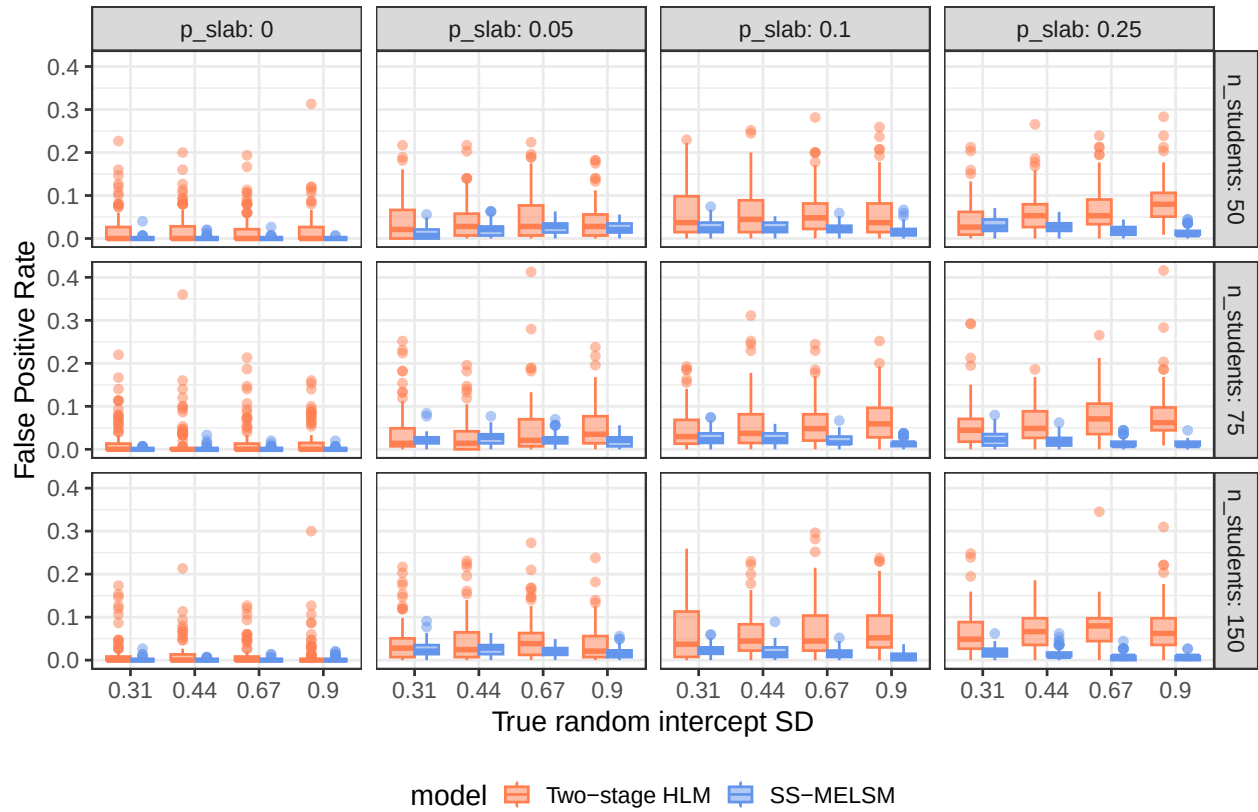
True Positive Rate

TPR for 150 schools

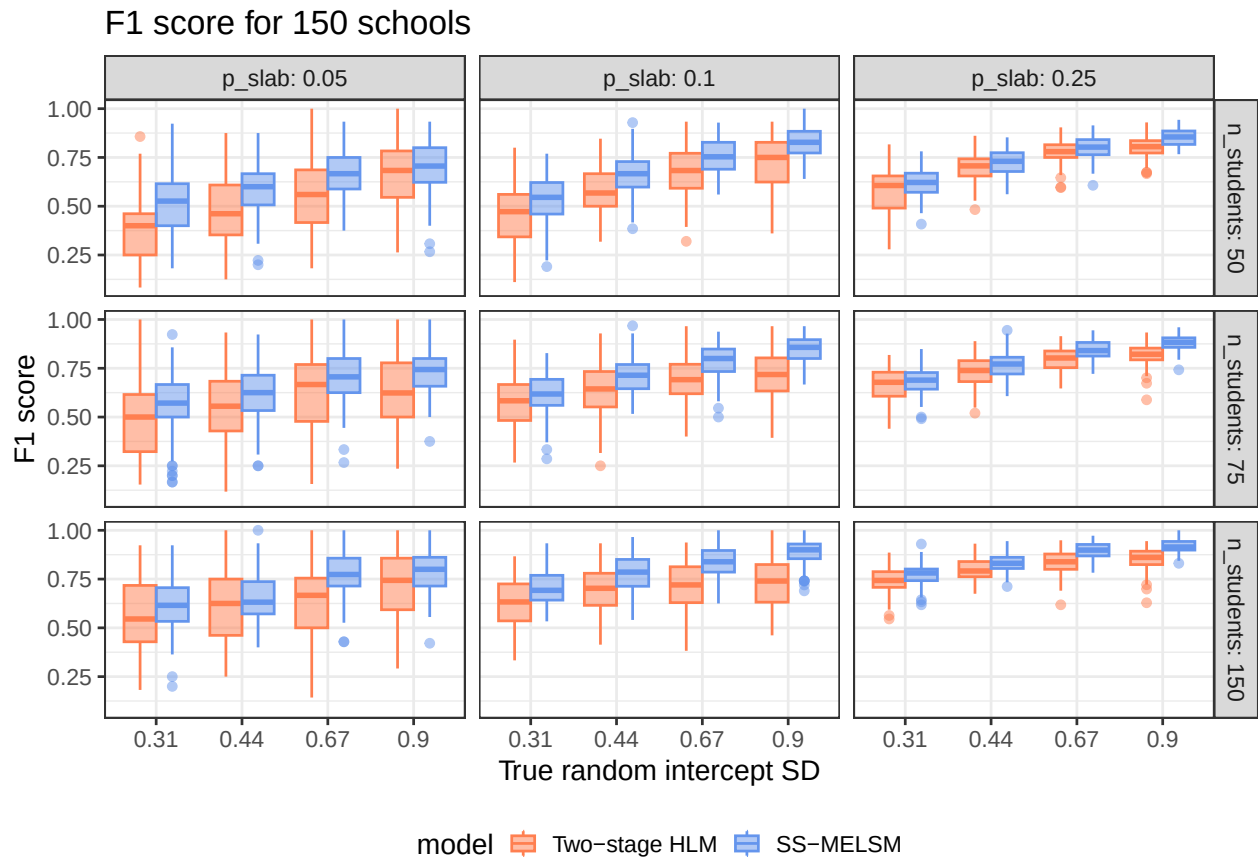


False Positive Rate

FPR for 150 schools

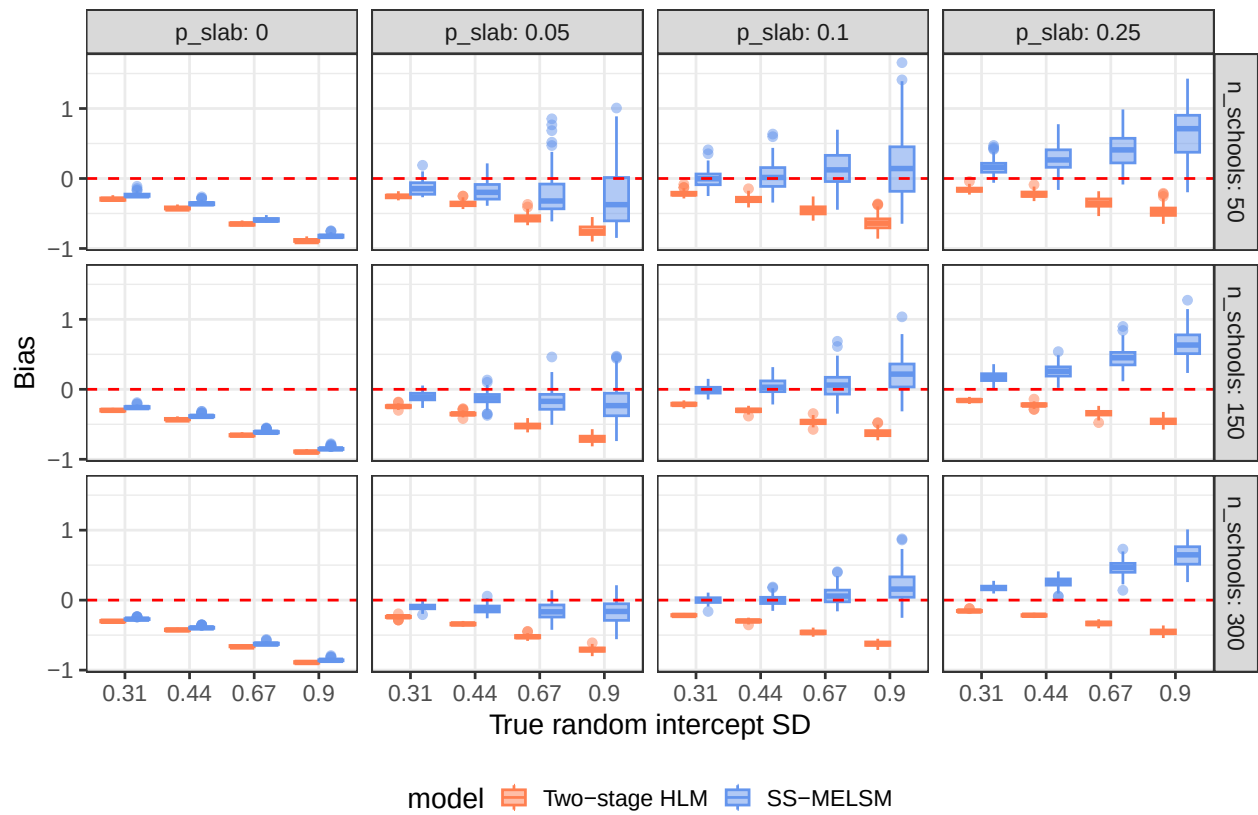


F1 score



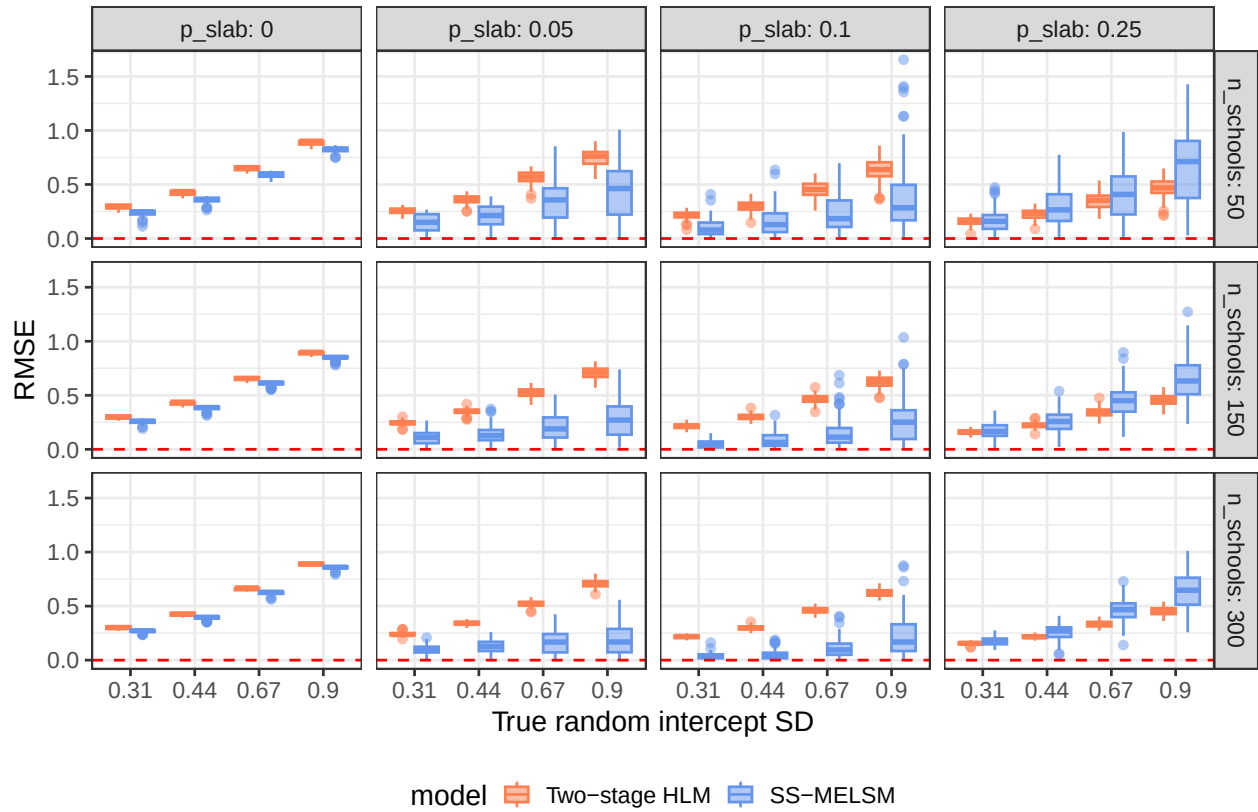
Bias

Bias for 75 students



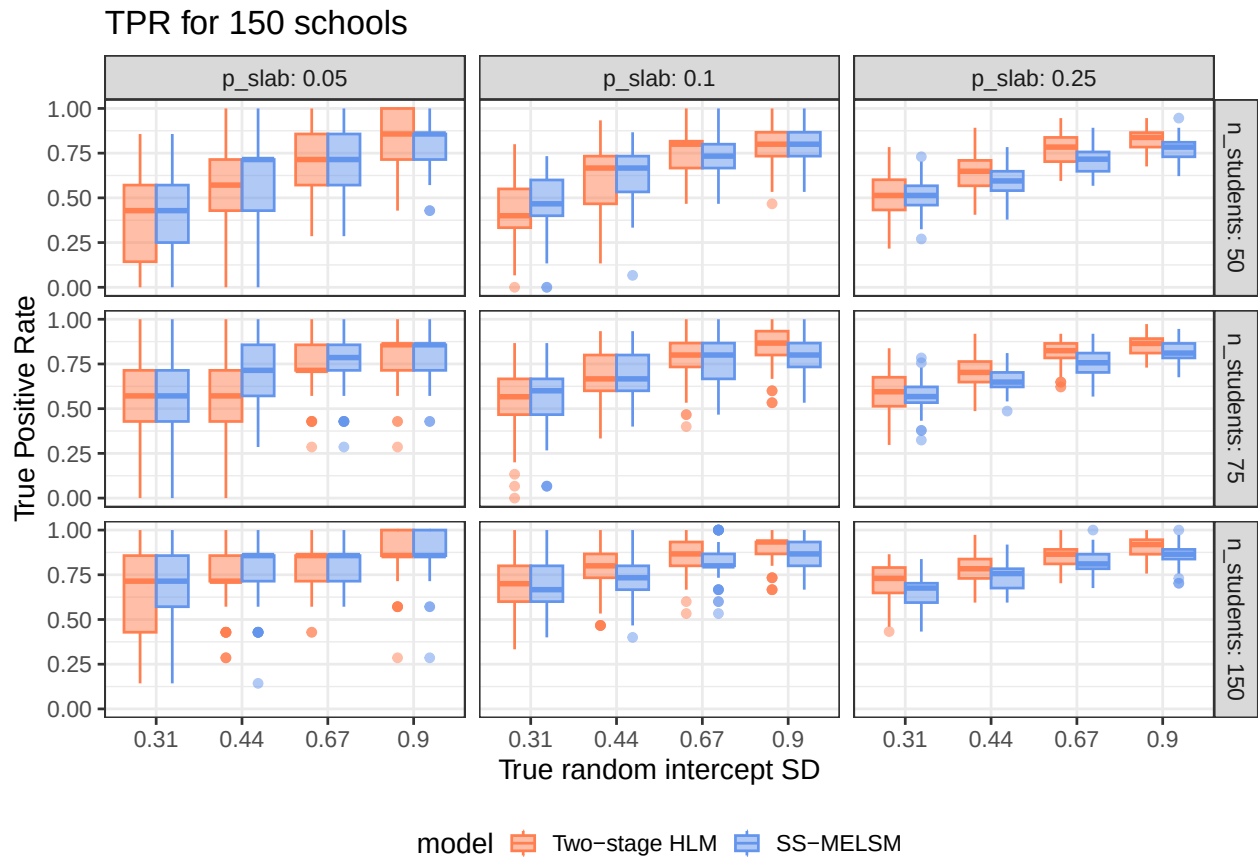
RMSE

RMSE for 75 students

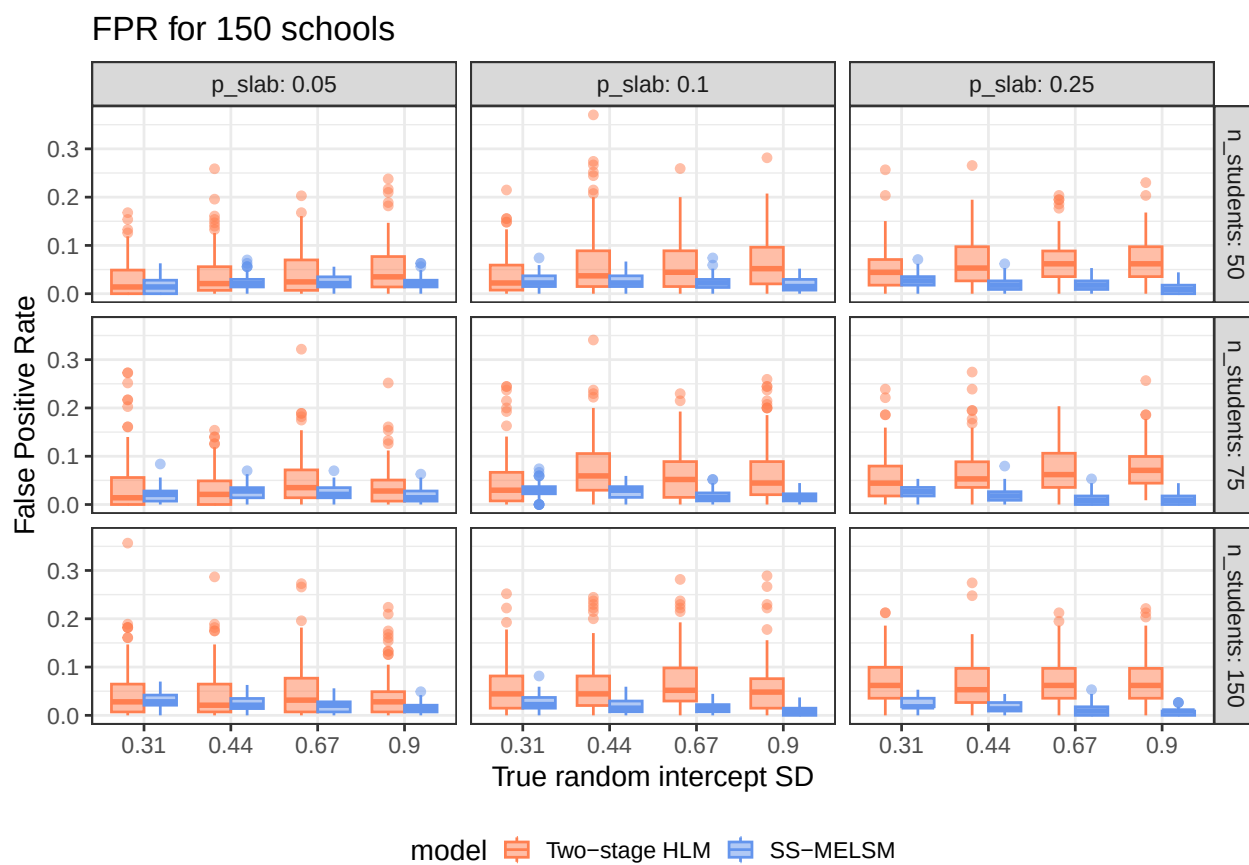


Correlated random effects ($\rho_{u_0t_0} = 0.5$)

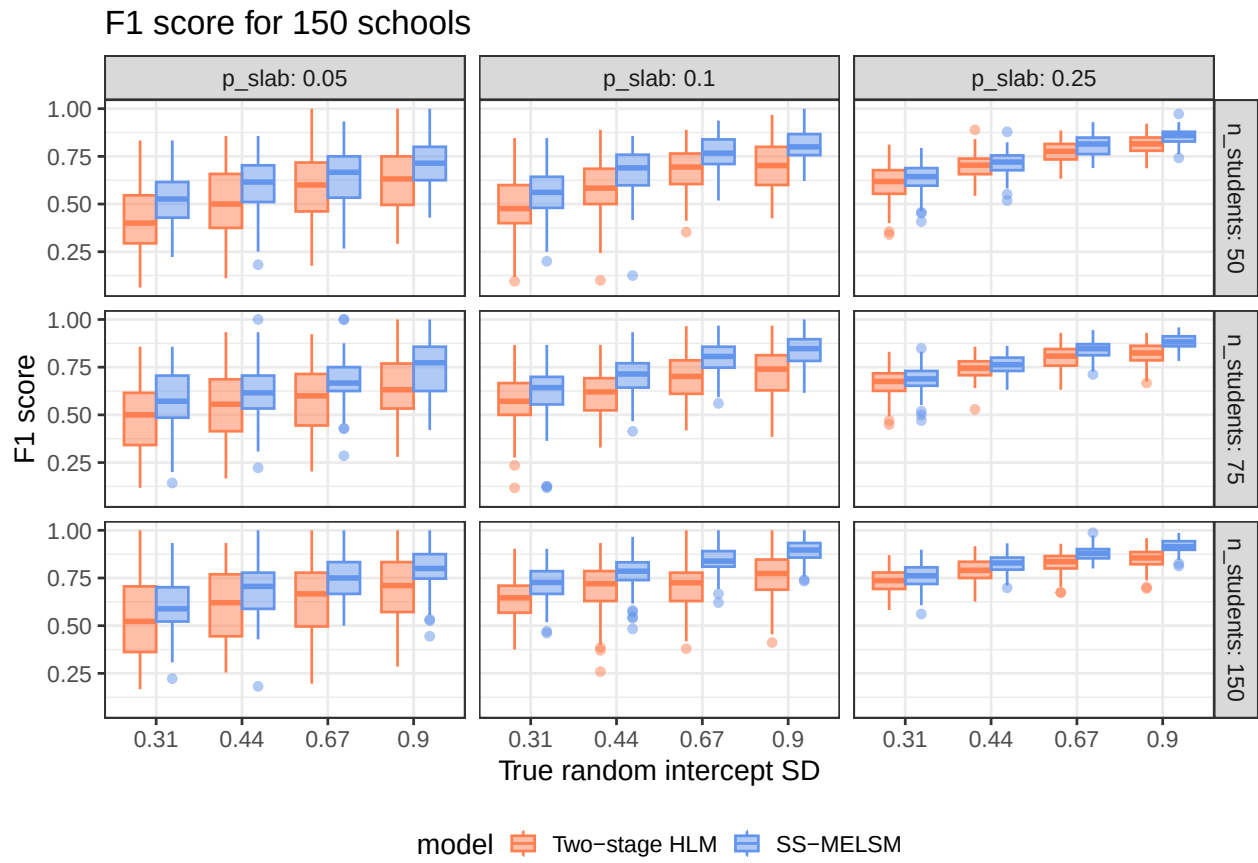
True positive rate



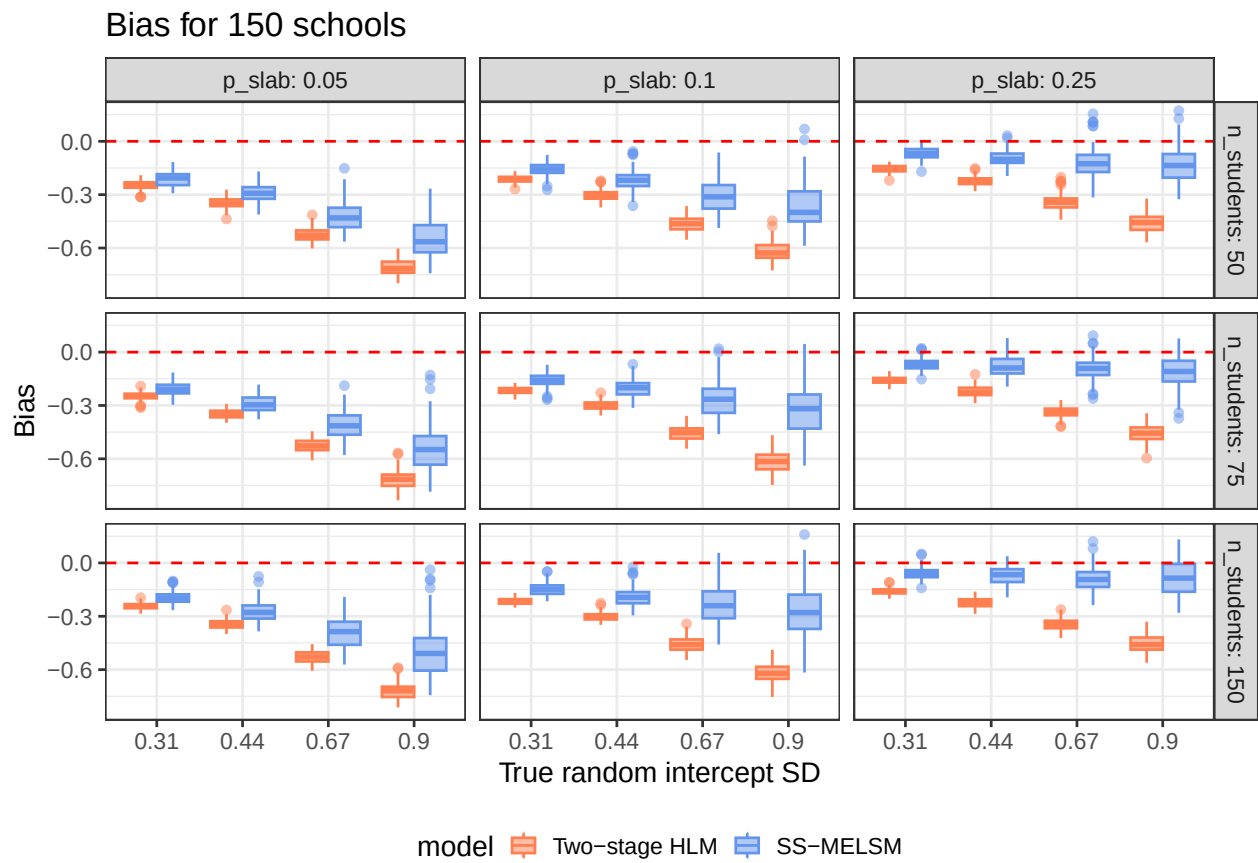
False positive rate



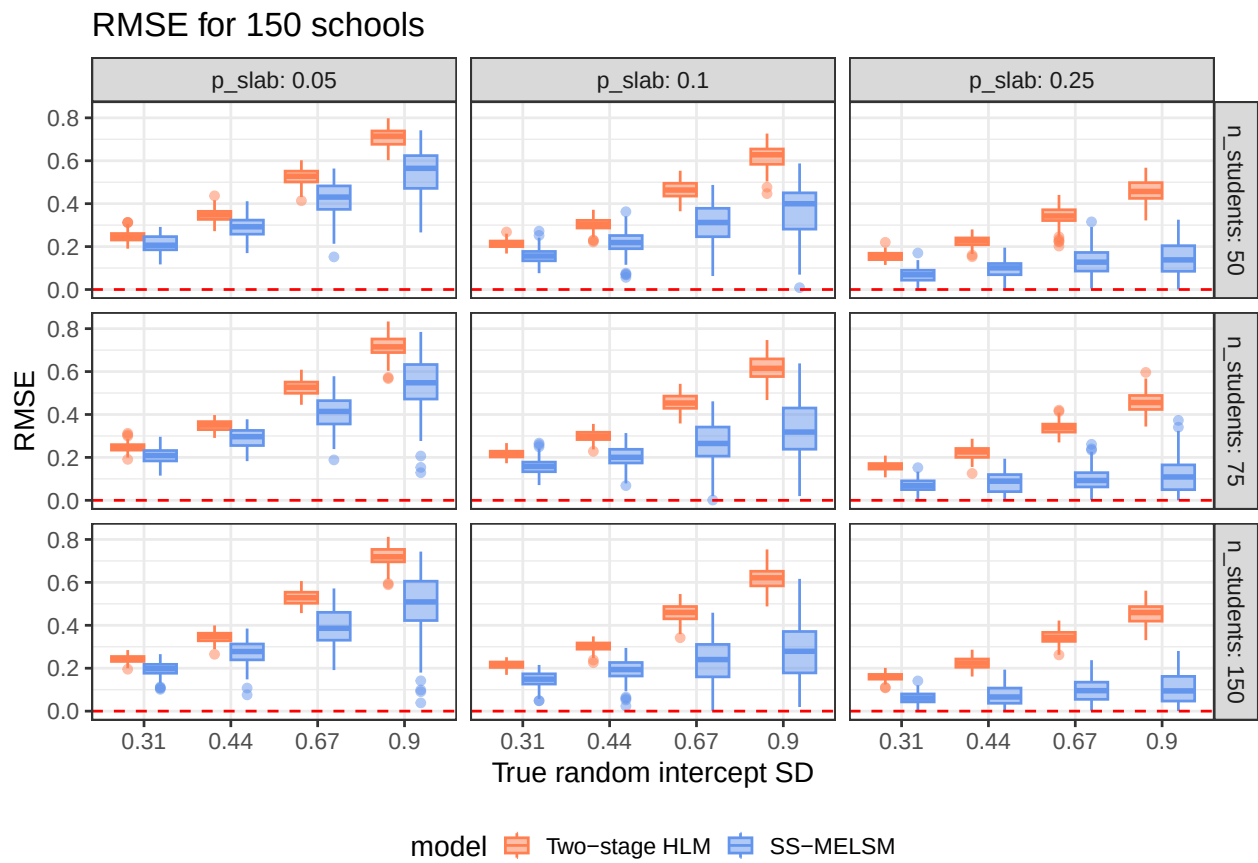
F1 score



Bias



RMSE



Estimated correlation coefficient

Correlation between location and scale random effects
for 150 schools

True correlation of 0.5

